

ERRACHIDIA MOULAY ALI CHERIF, Morocco

WMO: 602100

Lat: 31.948N

Lon: 4.398W

Elev: 3428

StdP: 12.96

Time Zone: 1.00 (E01)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	33.7	36.0	3.8	7.6	71.0	8.8	9.8	70.5	24.6	50.6	20.5	52.9	1.3	20	0.420

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.2	104.1	64.5	102.5	64.6	100.7	64.5	72.6	95.4	70.6	94.0	69.0	92.7	10.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.3	106.3	86.2	62.9	97.4	82.3	60.7	90.0	80.6	38.7	95.1	36.7	94.5	35.2	93.0	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.3	21.8	17.9	29.7	106.6	3.1	1.7	27.5	107.8	25.7	108.7	24.0	109.7	21.8	110.9
			27.0	73.3	3.5	4.2	24.5	76.3	22.5	78.8	20.5	81.2	18.0	84.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.9	48.9	52.8	60.3	67.4	75.3	83.7	90.9	88.3	79.7	69.5	57.9	50.5	(6)
(7)		DBStd	15.12	4.62	4.67	5.51	5.51	6.48	4.92	3.57	4.16	5.42	4.73	5.60	4.28	(7)
(8)		HDD50	153	77	23	2	0	0	0	0	0	0	5	46		(8)
(9)		HDD65	1737	500	343	163	38	6	0	0	0	16	221	450		(9)
(10)		CDD50	7035	42	101	322	521	784	1012	1269	1187	891	603	241	62	(10)
(11)		CDD65	3142	0	0	18	109	325	562	804	722	441	154	7	0	(11)
(12)		CDH74	44339	1	6	222	1039	3778	7985	13423	11082	5446	1279	77	1	(12)
(13)		CDH80	25718	0	0	25	271	1711	4600	9114	6979	2729	286	3	0	(13)
(14)	Wind	WSAvg	6.1	3.4	4.6	6.6	7.8	8.5	7.9	8.0	7.7	6.8	4.9	3.9	3.1	(14)
(15)	Precipitation	PrecAvg	5.4	0.3	0.5	0.5	0.4	0.4	0.2	0.1	0.2	0.7	0.9	0.8	0.4	(15)
(16)		PrecMax	11.0	1.3	1.9	1.7	2.6	1.8	1.1	0.8	3.0	4.2	6.8	2.7	(16)	
(17)		PrecMin	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
(18)		PrecStd	2.3	0.4	0.6	0.5	0.6	0.5	0.4	0.2	0.7	1.1	1.5	0.6	(18)	
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.7	74.0	83.2	89.6	97.6	103.5	106.9	105.1	99.7	89.1	80.1	72.0	(19)
(20)		MCWB	50.4	51.2	55.1	55.7	58.4	63.7	61.8	65.3	62.7	58.7	56.2	50.2	(20)	
(21)		2%	DB	66.9	70.9	79.5	86.2	94.8	100.8	105.2	102.9	97.0	85.7	76.2	68.0	(21)
(22)		MCWB	49.2	50.6	53.7	55.7	59.6	62.3	63.5	65.4	62.8	60.2	56.6	49.3	(22)	
(23)		5%	DB	64.0	68.1	76.4	83.2	91.8	98.8	103.7	101.3	94.7	83.0	73.3	64.7	(23)
(24)		MCWB	47.7	49.5	52.7	55.4	59.6	62.3	64.5	65.6	62.9	59.7	54.8	49.7	(24)	
(25)		10%	DB	60.9	65.3	73.5	80.1	89.1	96.3	102.1	99.2	92.1	80.8	70.3	61.6	(25)
(26)		MCWB	46.0	48.5	52.0	54.9	59.4	61.6	64.5	65.9	63.1	59.2	53.0	48.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	56.2	60.1	63.1	68.0	71.1	75.3	75.8	71.7	67.4	62.4	55.3	(27)
(28)		MCD B	66.8	68.5	77.2	79.1	87.8	94.7	100.7	99.0	90.7	79.3	75.9	64.7	(28)	
(29)		2%	WB	51.4	53.8	57.5	60.9	65.6	68.5	72.3	73.1	69.3	64.8	59.0	53.5	(29)
(30)		MCD B	63.0	66.1	74.3	77.7	85.5	93.2	98.1	95.0	88.4	78.5	70.6	62.0	(30)	
(31)		5%	WB	49.4	52.0	55.7	59.0	63.9	66.7	70.2	71.1	67.5	63.2	56.6	51.8	(31)
(32)		MCD B	60.7	63.6	71.7	75.6	83.1	90.8	97.2	92.9	86.4	76.6	70.2	60.5	(32)	
(33)		10%	WB	47.4	50.1	53.7	57.2	62.1	65.0	68.3	69.3	66.1	61.8	54.8	49.8	(33)
(34)		MCD B	58.4	61.5	69.3	73.7	81.3	88.4	95.8	92.2	84.8	75.0	67.2	59.2	(34)	
(35)	Mean Daily Temperature Range		MDBR	22.3	21.4	21.6	21.2	21.3	22.0	22.2	21.0	21.1	20.6	21.4	21.4	(35)
(36)		5% DB	MCD BR	27.4	26.9	25.3	24.7	24.0	24.3	23.8	22.4	23.5	22.8	24.4	26.9	(36)
(37)			MCWBR	14.4	13.1	10.9	9.6	9.0	8.9	8.3	7.8	8.1	8.6	11.4	13.8	(37)
(38)			MCD BR	24.2	22.8	22.4	21.4	21.2	21.7	22.2	20.9	20.9	20.6	22.7	22.7	(38)
(39)		5% WB	MCWBR	14.2	13.2	11.9	10.9	9.8	9.4	10.3	9.7	9.6	10.0	12.1	13.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.267	0.308	0.372	0.399	0.446	0.489	0.566	0.527	0.452	0.395	0.320	0.282	(40)	
(41)		taud	2.537	2.403	2.194	2.133	2.012	1.901	1.722	1.828	2.026	2.184	2.428	2.514	(41)	
(42)		Ebn at Noon	309	302	287	283	269	257	236	244	258	266	284	296	(42)	
(43)		Edh at Noon	26	33	44	49	56	62	74	66	52	41	29	26	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1199	1499	1869	2199	2380	2529	2406	2184	1867	1511	1228	1087	(44)	
(45)		RadStd	72	89	97	100	96	100	68	76	99	77	78	59	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page